

Multiple Choice (10 marks)

1. Which of the following lasers utilizes transitions that involve the energy levels of free atoms?
 - (a) Diode laser
 - (b) Dye laser
 - (c) Free-electron laser
 - (d) Gas laser
2. A refracting telescope consists of two converging lenses separated by 100 cm. The eyepiece lens has a focal length of 20 cm. The angular magnification of the telescope is
 - (a) 4
 - (b) 5
 - (c) 6
 - (d) 20
3. A resistor in a circuit dissipates energy at a rate of 1 W. If the voltage across the resistor is doubled, what will be the new rate of energy dissipation?
 - (a) 0.25 W
 - (b) 0.5 W
 - (c) 1 W
 - (d) 4 W
4. A single-electron atom has the electron in the $l = 2$ state. The number of allowed values of the quantum number m_l is
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 5
5. For which of the following thermodynamic processes is the increase in the internal energy of an ideal gas equal to the heat added to the gas?
 - (a) Constant temperature
 - (b) Constant volume
 - (c) Constant pressure
 - (d) Adiabatic

True / False (10 marks)

Answer True or False

6. True or False: In a superconductor, the attraction between Cooper pairs arises from interactions with the ionic lattice.
7. True or False: When 100 photons are sent to a detector with a quantum efficiency of 0.1, it will detect an average of 10 photons with an rms deviation of about 3.
8. True or False: An electron in the $n = 4, l = 1$ state can transition to the $n = 3, l = 2$ state without violating selection rules.
9. True or False: The speed of light in a nonmagnetic dielectric material with a dielectric constant of 4.0 is 1.5×10^8 m/s.
10. True or False: The Sun's energy is primarily produced by the fusion of two hydrogen atoms into one helium atom, resulting in significant energy release.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how Fermat's principle of ray optics, which states that light follows the path requiring the least time, can be used to derive Snell's law of refraction and the law of reflection.
12. Discuss the work required to accelerate a proton from rest to a speed of $0.6c$. Include calculations and analyze the implications of relativistic effects on this process.

End of Paper